

Immune gene and protein profiles across blood, cerebrospinal fluid and peripheral nerve in inflammatory neuropathies

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Abstract

Background

Blood, cerebrospinal fluid (CSF) and peripheral nerve provide complementary views of immune changes in inflammatory neuropathies. We characterised immune gene and protein profiles in Guillain–Barré syndrome (GBS) and chronic inflammatory demyelinating polyradiculoneuropathy (CIDP) using public transcriptomic datasets and published proteomic and genetic findings. Gene modules were compared across three acute GBS blood cohorts, and published CSF protein coefficients were pooled across two cohorts. Nerve expression was analysed at the donor level in nine individuals with CIDP and 11 with chronic idiopathic axonal polyneuropathy, with published genetic candidates mapped to nerve cell populations.

21 **Results**

22 CXCL8 recruitment and complement scores were higher in acute GBS across all three blood
23 cohorts, with pooled Hedges g estimates of 1.08 (95% confidence interval -0.63 to 2.78) and
24 1.11 (-1.73 to 3.95), respectively. Nine of ten selected CSF proteins had positive coefficients
25 in both cohorts and pooled $q < 0.05$ under the fixed effect model, whereas CD5 showed
26 opposite directions between cohorts (pooled $q = 0.984$). In nerve macrophages, the Fc
27 receptor module score was lower in CIDP than in the axonal disease comparator, with $q =$
28 0.0087 after correction across eight macrophage modules and $q = 0.087$ across all 80 nerve
29 module comparisons. This difference persisted after adjustment for age and centre but was
30 attenuated after adjustment for macrophage composition. Correlations among nerve modules
31 and with disability did not survive multiple testing correction (all $q > 0.05$). Among the
32 genetic candidates, *CDH4* expression was highest in Schwann cell states, whereas *GNG7*
33 expression was highest in B cells.

34 **Conclusions**

35 These findings highlight recurring recruitment and complement expression patterns in acute
36 GBS blood, inflammatory protein changes in CSF, and distinct cellular expression patterns in
37 CIDP nerve. Interpretation of the macrophage Fc receptor difference depends on cellular
38 composition and the scope of multiple testing correction. Genetic candidate localisation
39 provides a cellular context for published CIDP associations.

40 **Keywords**

41 Inflammatory neuropathy; transcriptomics; proteomics; blood–nerve barrier; macrophages;
42 genetic susceptibility

43 **Background**

44 Guillain–Barré syndrome (GBS) and chronic inflammatory demyelinating
45 polyradiculoneuropathy (CIDP) are inflammatory neuropathies with distinct clinical courses
46 and diagnostic criteria [1,2,3]. Antibodies, complement and macrophages contribute to
47 peripheral nerve injury, although their relative roles vary across diseases and subtypes
48 [4,5,6,7,8,9]. Peripheral nerve atlases and experimental studies have further characterised
49 diverse immune and Schwann cell responses to inflammation and injury [10,11,12,13]. These
50 responses extend to processes involved in nerve repair, as illustrated by the effects of
51 leukaemia inhibitory factor (LIF) on Schwann cell survival, proliferation and migration in
52 experimental systems [14,15]. Examining immune and repair gene expression together may
53 therefore provide a broader view of the molecular changes associated with inflammatory
54 neuropathies.

55 The blood–nerve barrier (BNB) provides an anatomical interface between circulating immune
56 cells and the peripheral nerve environment. It regulates exchange between the circulation and
57 the endoneurium, with endothelial junctions and associated mural cells helping maintain its
58 specialised properties [16,17,18]. Experimental studies have implicated leukocyte integrins,
59 endothelial adhesion molecules and chemokines in immune cell trafficking across this
60 interface [19,20,21]. Altered junctional protein expression in CIDP nerve further supports
61 examining the BNB in the context of local inflammation [22].

62 Molecular studies of blood, cerebrospinal fluid (CSF) and peripheral nerve offer
63 complementary perspectives on these processes. Blood transcriptomic studies using whole
64 blood, leukocytes and peripheral blood mononuclear cells (PBMCs) characterise systemic
65 immune responses in GBS [23,24,25]. CSF proteomics provides information on inflammation
66 and molecular injury, while nerve transcriptomics identifies the cell populations expressing
67 related genes [26,27]. Additional studies have described monocyte changes in acute

inflammatory demyelinating polyradiculoneuropathy (AIDP), altered peripheral immune profiles in CIDP, and associations between blood gene expression and treatment response [28,29,30]. Experiments using patient immunoglobulin further implicate the BNB in inflammatory neuropathy [31], and genetic studies nominate candidate genes for investigation [32]. Considering these resources together offers an opportunity to examine how systemic immune expression relates to the pathways and cell populations represented in peripheral nerve.

Here, we compared immune gene modules across acute GBS blood cohorts, synthesised selected published CSF protein findings, and examined immune and repair modules in CIDP nerve. We also mapped published CIDP genetic candidates to nerve cell populations to establish their cellular expression context. These comparisons drew on separate participant cohorts to identify recurring biological themes, with differences in diagnosis, specimen type and assay informing the interpretation of patterns across datasets.

Methods

Study design and datasets

We examined immune profiles across blood, CSF and peripheral nerve using human transcriptomic datasets and published proteomic and genetic findings. GSE211225 provided six whole blood profiles from acute GBS, ten from the postacute phase and six from healthy controls [23]. GSE31014 included leukocyte microarrays from seven patients with GBS and seven healthy controls [24], and PRJNA1293757 provided PBMC transcriptomes from three untreated patients with AIDP and two healthy controls [25]. Blood module comparisons used the acute disease and healthy control groups from all three datasets.

Published CSF evidence included protein and endogenous peptide identifications by diagnostic group from the study associated with PXD002911 [26], together with reports on

92 complement, quantitative proteomics and inflammatory proteins [33,34,35]. For quantitative
93 synthesis, we extracted adjusted protein coefficients from the independent discovery and
94 replication cohorts reported by Kmezic et al. [36].

95 Peripheral nerve expression was examined using GSE285983, which comprised 365,708
96 nuclei from 37 sural nerves and included nine donors with CIDP and 11 with chronic
97 idiopathic axonal polyneuropathy (CIAP) [27]. The published human BNB transcriptome
98 associated with GSE107574 served as a reference for interpreting endothelial expression [17].
99 Additional tissue evidence came from the publication associated with PXD056286, which
100 reported histological observations from 55 nerve biopsies in CIDP and its variants, with
101 proteomic findings from a smaller subset [37]. To examine the cellular expression of genetic
102 candidates, we selected genes from the genetic association, Mendelian randomisation and
103 colocalisation results reported by Du et al. [32].

104 Table 1 summarises the resources and their analytical roles, and Additional file 1 provides
105 database names, accession identifiers, original filenames and source links.

106 **Gene modules**

107 We evaluated seven blood gene modules representing CXCL8 recruitment through CXCR1
108 and CXCR2, LIF and oncostatin M (OSM) signalling through gp130, complement, Fc
109 receptors, interferon signalling through JAK and STAT, inflammatory monocytes and
110 transendothelial migration. Nerve analyses used eight modules representing BNB identity,
111 leukocyte transmigration, CXCL8 recruitment, LIF and related gp130 signalling,
112 complement, Fc receptors, macrophage state and Schwann cell myelin and repair. Module
113 scores summarised expression across the corresponding gene sets, whose constituent genes
114 are specified in the analysis scripts. The BNB module combined endothelial and junctional
115 genes, while the Schwann cell module combined myelin and injury response genes. These

116 scores summarised gene expression rather than directly measuring barrier integrity or repair
117 capacity.

118 **Blood cohort analysis**

119 For PRJNA1293757, we assigned quality controlled PBMCs to broad lineages using marker
120 gene scores. We then generated monocyte pseudobulk profiles by summing counts within
121 each donor, normalising them to counts per million (CPM) and applying a $\log_2(\text{CPM} + 0.5)$
122 transformation. Within each blood cohort, we standardised gene expression across
123 participants and calculated module scores by averaging the standardised expression of
124 available genes in each module. Differences between acute GBS and healthy controls were
125 quantified using Hedges g , and mean module scores were compared using exact label
126 permutation tests. P values were adjusted across the seven modules within each cohort using
127 the Benjamini and Hochberg procedure [38].

128 Effects from whole blood, leukocytes and monocyte pseudobulk were pooled using a random
129 effects model, with variance between cohorts estimated by the DerSimonian and Laird
130 method and heterogeneity summarised by τ^2 and I^2 [39]. Confidence intervals were calculated
131 using the modified Hartung and Knapp method, with the residual variance scale constrained
132 to be at least one and a t distribution with degrees of freedom equal to the number of cohorts
133 minus one [40]. We assessed the influence of each cohort on the pooled estimates by
134 repeating the synthesis after omitting one cohort at a time. Postacute profiles contributed to
135 the original preprocessing but were excluded from the inferential comparisons reported here,
136 and the archived study design did not establish whether donors were independent across the
137 acute and postacute phases.

138 **CSF protein synthesis**

139 The quantitative CSF synthesis included the ten proteins with coefficients and standard errors
140 reported for both cohorts in Table 4B of Kmezic et al. [36]. These coefficients compared
141 GBS with healthy controls on the $\log_2$ normalised protein expression (NPX) scale and were
142 adjusted for age, sex and the CSF sample handling marker CCL19. All paired estimates in
143 this reported subset were included, regardless of their direction or statistical significance. For
144 this exploratory synthesis, we pooled coefficients using inverse variance weighting under a
145 fixed effect model, calculated 95% confidence intervals using a normal approximation, and
146 adjusted P values across the ten pooled tests using the Benjamini and Hochberg procedure.
147 DerSimonian and Laird random effects estimates were also reported alongside Q and I^2 to
148 assess heterogeneity.

149 For the descriptive synthesis, we summarised CSF protein concentrations, protein and peptide
150 identifications, and functional annotations from four reports [26,33,34,35]. We grouped
151 findings by biological category and counted the studies and assay types represented in each
152 category. Concentration measurements were grouped as one broad assay type.

153 **Nerve pseudobulk analysis**

154 We grouped GSE285983 nuclei by donor and annotated cell state. The groups included
155 macrophages, T and natural killer cells, B cells, granulocytes, endoneurial endothelial cells,
156 other endothelial cells, pericytes, perineurial cells, endoneurial stroma, and Schwann cells in
157 myelinating, nonmyelinating, and repair and damage states. Groups with at least 20 nuclei per
158 donor were eligible. We summed counts within each donor and cell group, calculated counts
159 per million (CPM) using the total counts across all genes, and transformed expression as
160 $\log_2(\text{CPM} + 0.5)$. Comparisons between CIDP and CIAP used donors as independent
161 observations [41,42,43].

162 **Nerve modules and macrophage sensitivity analyses**

163 Within each cell group, we standardised each gene across eligible donors from all diagnostic
164 categories and averaged the available gene z scores to obtain module scores. We compared
165 CIDP with CIAP using the Mann and Whitney U test and expressed the difference as the
166 mean CIDP score minus the mean CIAP score. Multiple testing correction covered eight
167 modules within each cell group. We also applied correction across all 80 eligible nerve
168 module comparisons. Cell fractions were compared separately at donor level, with correction
169 across cell groups. These fractions describe the relative representation of each population in
170 the sampled nuclei.

171 The macrophage Fc receptor module contained 11 measured genes. *FCGR2C*, the twelfth
172 specified gene, was absent from the expression matrix. We assessed individual genes using
173 the Mann and Whitney U test with correction across 11 tests, and four functional summaries
174 with a separate correction. The q values for these comparisons without covariate adjustment
175 were derived from the rank tests, whereas confidence intervals for mean differences were
176 obtained from separate HC3 models. Sensitivity analyses restricted the comparison to male
177 donors, standardised genes within the 20 CIDP and CIAP donors, adjusted for age and centre,
178 and added sex to the adjusted model. Linear models used HC3 covariance estimates and
179 residual degrees of freedom for confidence intervals. An exact permutation test preserved the
180 number of CIDP donors within each centre. Additional file 2 provides the model
181 specifications.

182 We matched the original macrophage annotations to refined immune annotations using
183 nucleus barcodes. Composition summaries retained nuclei with unmatched labels and those
184 reassigned to other categories. We represented donor composition using centred log ratios
185 [44] and added its first two principal components to the model containing age and centre. We
186 also reaggregated raw counts for all 20 CIDP and CIAP donors under alternative macrophage

187 definitions. Tests of refined subtypes required at least 20 nuclei per donor and at least three
188 eligible donors in each diagnosis group. Multiple testing correction covered the nine eligible
189 refined subtypes.

190 **Nerve module and disability correlations**

191 We paired five module scores within each of the nine CIDP donors. These were endothelial
192 LIF, endothelial leukocyte transmigration, macrophage complement, macrophage Fc
193 receptors, and myelin and repair in Schwann cells annotated as repair and damage states.
194 Spearman correlations described their ten pairwise relationships. We also tested 15
195 combinations of cell group and module for associations with Inflammatory Neuropathy
196 Cause and Treatment (INCAT) disability scores. Multiple testing correction was applied
197 separately to the ten intermodule correlations and the 15 disability correlations.

198 **Comparison across compartments**

199 We compared recurring biological programmes across compartments. A programme entered
200 the evidence map if it had a consistent effect direction across blood cohorts, a corresponding
201 published CSF protein or peptide annotation, or expression in a relevant nerve cell
202 population. Each cell identifies the evidence type, and statistical comparisons retain their own
203 testing families. For nerve interferon related genes, we extracted cell cluster marker results
204 from Supplementary Data 3 and polyneuropathy versus control results from Supplementary
205 Data 9 of Heming et al. [27]. The source controls were residual sural nerve autograft tissue
206 from patients with traumatic nerve injuries. The map compares biological themes across
207 independently sampled cohorts without calculating a single combined score.

208 **Genetic candidate localisation**

209 We selected *CDH4*, *DIRAS1*, *GNG7* and *SLC39A3* from the genetic study of 516 CIDP cases
210 and 403,545 controls [32]. Candidate selection used the *CDH4* locus reported in women and

211 the published Mendelian randomisation and colocalisation findings for the other three genes.
212 We examined their expression across nerve cell groups and compared CIDP with CIAP.
213 CellBender counts from GSE285983 were normalised to 10,000 counts per nucleus,
214 transformed as $\log_2(\text{CP10k} + 1)$, and summarised by donor and cell group. We calculated
215 mean expression and the percentage of nuclei with detectable expression for each donor.
216 Figure 5B shows the arithmetic mean of each measure across the nine CIDP donors, with
217 equal weight given to each donor. For each gene and cell group, 5,000 bootstrap resamples of
218 eligible CIDP donors provided intervals around mean expression, using the random seed
219 20260904. Differences between CIDP and CIAP were tested using the Mann and Whitney U
220 test, with P values adjusted across the 11 genes in the archived genetic candidate screen
221 within each cell group, including the four CIDP candidates reported here.

222 **Statistical analysis**

223 Participants and nerve donors were the units of statistical inference. All disease contrasts and
224 correlation tests were two sided. We denote P values adjusted by the Benjamini and
225 Hochberg procedure as q and use $q < 0.05$ within each stated testing family. The Fc
226 component and sensitivity analyses were planned after the initial macrophage association had
227 been observed and are exploratory. Additional file 2 describes the inputs and intermediate
228 tables needed to reproduce each component. Additional file 3 contains Supplementary
229 Figures 1 to 4, which present candidate gene expression, published nerve observations and
230 macrophage sensitivity analyses.

231 **Results**

232 **Datasets and analytical roles**

233 The three blood cohorts provided estimates from whole blood, leukocytes and monocytes.
234 Published CSF studies contributed protein measurements, and the sural nerve atlas provided

expression profiles by donor and cell group. The BNB reference and genetic findings supported the interpretation and selection of nerve genes (Fig. 1 and Table 1).

Immune gene modules in acute GBS blood

Acute GBS was associated with higher scores for six of the seven modules in GSE211225 after multiple testing correction. These were *CXCL8* recruitment through CXCR1 and CXCR2 (Hedges $g = 1.25$, $q = 0.045$), *LIF* and *OSM* signalling through gp130 ($g = 2.19$, $q = 0.0076$), complement ($g = 2.49$, $q = 0.0076$), Fc receptors ($g = 1.36$, $q = 0.038$), interferon signalling through JAK and STAT ($g = 1.07$, $q = 0.045$), and inflammatory monocytes ($g = 1.93$, $q = 0.015$). The transendothelial migration score was also higher ($g = 0.92$, $q = 0.126$). These results indicate higher expression across several immune gene sets in acute GBS whole blood.

Six of the seven module effects were positive in the independent leukocyte cohort GSE31014. *CXCL8* recruitment, complement, Fc receptors, transendothelial migration, interferon signalling and inflammatory monocytes changed in the same direction as in whole blood. In PRJNA1293757 monocytes, *CXCL8* recruitment had the largest positive effect ($g = 1.97$), followed by a positive complement effect. The remaining module effects varied in direction.

CXCL8 recruitment and complement had positive effects in all three cohorts. The pooled Hedges g was 1.08 for *CXCL8* recruitment (95% confidence interval -0.63 to 2.78 , $P = 0.114$, $I^2 = 0\%$) and 1.11 for complement (95% confidence interval -1.73 to 3.95 , $P = 0.234$, $I^2 = 57.2\%$). All seven pooled confidence intervals included zero. Four other modules had positive effects in two cohorts, while *LIF* and *OSM* signalling was positive only in whole blood (Fig. 2 and Table 2).

Omitting each cohort in turn left the pooled direction positive for CXCL8 recruitment ($g = 0.97$ to 1.41) and complement ($g = 0.48$ to 1.55). Both individual effects in each retained pair were positive, although the intervals based on two cohorts were wide. Every other module had at least one cohort with a discordant effect direction.

Heterogeneity differed across modules. I^2 was 0% for CXCL8 recruitment and interferon signalling, 1.1% for transendothelial migration, 49.5% for inflammatory monocytes, 57.2% for complement, 72.7% for Fc receptors and 81.2% for LIF and OSM signalling. The largest variation therefore concerned pathways with strong effects in whole blood and smaller or opposite effects in the other blood populations. With three cohorts, platform, cell composition and clinical differences remain overlapping explanations for this variation.

Published CSF protein findings

Nine of the ten selected CSF proteins (SELE, IL2RA, CCL3, CR2, CD1C, THBD, NRP1, CD38 and IL6) had positive coefficients in both cohorts and pooled q values below 0.05 under the fixed effect model after correction across all ten proteins. Within this subset, SELE and IL2RA were the only proteins that met the adjusted significance threshold in both cohorts in the source study [36]. CD5 showed opposite directions in the discovery and replication cohorts, with coefficients of -0.0882 and 1.51 , respectively, yielding a pooled coefficient of 0.0024 (95% confidence interval -0.237 to 0.242 , $q = 0.984$). Individual and pooled coefficients are presented in Figure 3A and B. Heterogeneity was greatest for IL2RA ($I^2 = 94.9\%$), CD5 ($I^2 = 89.0\%$), IL6 ($I^2 = 80.7\%$) and THBD ($I^2 = 79.5\%$). Under the DerSimonian and Laird random effects model, the 95% confidence intervals for IL2RA, NRP1 and IL6 included zero, indicating sensitivity to the pooling model.

Four additional CSF reports provided complementary descriptive evidence. The study associated with PXD002911 reported protein and peptide identifications with functional

282 annotations [26]. D'Aguanno et al. described complement and inflammatory proteins [33],
283 Ding et al. reported proteins related to protein handling and neural structure [34], and Sun et
284 al. examined inflammatory proteins alongside routine CSF measurements [35]. Across these
285 reports, local inflammation was represented in three. Protein accumulation was represented in
286 two reports by elevated total CSF protein concentrations [26,35]. Structural proteolysis,
287 complement, and protein handling and neural structure were each represented in one report
288 (Fig. 3C).

289 **Immune and repair gene expression in CIDP nerve**

290 The sural nerve atlas revealed distinct cellular distributions of immune and repair gene
291 expression. Macrophages expressed *C3*, *C3AR1* and *FCGR2A*, while endoneurial endothelial
292 cells expressed *CLDN5*, *ICAM1*, *VCAM1*, *LIFR*, *IL6ST* and *OSMR*. Schwann cell states
293 expressed genes encoding gp130 receptor components alongside genes involved in
294 myelination, stress responses and repair (Fig. 4A). The expression of *CLDN5* and adhesion
295 genes in endothelial cells was consistent with findings from the published human BNB
296 transcriptome [17].

297 The macrophage Fc receptor module showed the strongest statistical evidence of a difference
298 between CIDP and CIAP, with a mean module score difference of -0.673 (CIDP minus
299 CIAP, $P = 0.0011$). The corresponding q value was 0.0087 after correction across eight
300 macrophage modules and 0.087 after correction across all 80 eligible nerve module
301 comparisons. Macrophage leukocyte transmigration and macrophage state scores were also
302 lower in CIDP, with differences of -0.345 ($P = 0.0185$, $q = 0.074$) and -0.368 ($P = 0.0334$, q
303 $= 0.089$), respectively. In perineurial cells, the macrophage state and complement module
304 scores were higher in CIDP by 0.758 and 0.348 , respectively (both $q = 0.109$). The
305 endothelial BNB identity score was lower by 0.404 ($P = 0.0276$, $q = 0.221$). These
306 comparisons are shown in Figure 4B.

307 All nine CIDP donors were male, compared with seven of the 11 CIAP donors, and the Fc
308 receptor score remained lower in CIDP when the comparison was restricted to male donors
309 (difference -0.687 , $P = 0.0007$). Adjustment for age and centre yielded a similar estimate
310 (difference -0.690 , 95% confidence interval -1.040 to -0.339 , $P = 0.0008$), and an exact
311 permutation test within centres gave $P = 0.0048$. Estimates remained negative after individual
312 donors, centres or module genes were omitted in turn (Supplementary Fig. 3).

313 At the gene level, four of the 11 measured genes, *FCGR2A*, *FCER1G*, *TYROBP* and *HCK*,
314 showed lower expression in CIDP with $q < 0.05$ in the rank based comparisons. The q value
315 for *FCGRT* was 0.713 , while *FCGR1A* had a positive estimate with $q = 0.148$. The summary
316 score for signalling components was lower ($q = 0.039$), whereas the activating receptor,
317 inhibitory receptor and neonatal Fc receptor summaries all had q values above 0.05 . The
318 lower composite score therefore reflected differences in particular receptor and signalling
319 genes.

320 Adding two macrophage composition components to the model containing age and centre
321 attenuated the estimated difference to -0.310 (95% confidence interval -1.038 to 0.419 , $P =$
322 0.375). A lower score was also observed in the combined group of refined macrophage
323 subtypes ($P = 0.0062$), although all nine eligible tests of individual refined subtypes and all
324 32 annotation fraction tests had q values above 0.05 (Supplementary Fig. 4).

325 Across the broad nerve groups, no cell fraction comparisons met the threshold after multiple
326 testing correction (all $q > 0.05$). The largest numerical increases in CIDP were observed in
327 nonmyelinating Schwann cells (0.064 , $q = 0.515$), endoneurial stroma (0.027 , $q = 0.515$) and
328 perineurial cells (0.024 , $q = 0.892$) (Fig. 4C).

329 Among the nine CIDP donors, the strongest association with INCAT disability was observed
330 for the myelin and repair score in Schwann cells annotated as repair and damage states
331 (Spearman $\rho = 0.840$, $P = 0.0046$, $q = 0.069$). Endothelial LIF scores showed a weaker

332 positive association ($\rho = 0.602$, $P = 0.086$), but none of the 15 disability correlations met the
333 threshold after multiple testing correction (all $q > 0.05$, Fig. 4D).

334 Published tissue findings provided additional context for these expression patterns. Stascheit
335 et al. reported terminal complement complex deposition on endoneurial capillaries in 52 of 55
336 biopsies from patients with CIDP or its variants [37], corresponding to 94.5% (calculated
337 Wilson 95% confidence interval 85.1% to 98.1%). The source study also described
338 macrophages expressing CD68 and T cells expressing CD8. In the proteomic subset of nine
339 CIDP cases and two controls, the authors reported higher C3, C5 and C9 abundance, whereas
340 C6 was elevated in only two cases. These published observations are summarised in
341 Supplementary Figure 2.

342 **Relationships among nerve modules**

343 Within the nine CIDP donors, endothelial LIF scores correlated positively with endothelial
344 leukocyte transmigration scores ($\rho = 0.667$, $P = 0.0499$, $q = 0.166$) and Schwann cell myelin
345 and repair scores ($\rho = 0.783$, $P = 0.0125$, $q = 0.125$). Endothelial transmigration scores also
346 correlated positively with macrophage complement scores ($\rho = 0.683$, $P = 0.0424$, $q = 0.166$),
347 although none of the ten intermodule correlations met the threshold after multiple testing
348 correction (all $q > 0.05$).

349 The evidence map placed these nerve findings alongside the blood and CSF results (Fig. 5
350 and Table 3). CXCL8 recruitment and complement showed consistent effect directions across
351 blood cohorts, with related inflammatory or complement annotations represented in CSF. Fc
352 receptor module scores were higher in acute GBS whole blood than in healthy controls but
353 lower in CIDP nerve macrophages than in CIAP macrophages. Within nerve, the expression
354 of adhesion genes in endothelial cells and repair genes in Schwann cells defined the cellular
355 context of these programmes, while CSF studies described protein accumulation and

356 proteolysis. Published source atlas markers showed *OAS1* enrichment in macrophages and
357 *MX1* enrichment in arterial endothelial cells, while the source polyneuropathy versus control
358 comparison showed lower *MX1* expression in nonmyelinating Schwann cells [27].

359 **Cellular expression of genetic candidates**

360 Four genetic candidates were selected from two forms of evidence reported by Du et al. [32].
361 *CDH4* was linked to the 20q13.33 locus reported in women, while *DIRAS1*, *GNG7* and
362 *SLC39A3* were selected from the Mendelian randomisation and colocalisation findings. The
363 published estimates underlying this selection are presented in Figure 5C.

364 Expression summaries across CIDP donors showed distinct cellular patterns among these
365 candidates. *CDH4* expression was highest in Schwann cells annotated as repair and damage
366 states and nonmyelinating Schwann cells. *GNG7* showed the highest expression in B cells,
367 with additional expression detected in macrophages and BNB endothelial cells. *SLC39A3*
368 expression was low overall, with the highest mean in BNB endothelial cells, whereas *DIRAS1*
369 was sparsely expressed (Fig. 5B). Bootstrap intervals summarised uncertainty in the
370 expression estimates for each cell group (Supplementary Fig. 1). None of the displayed
371 expression comparisons between CIDP and CIAP reached significance after multiple testing
372 correction (all $q > 0.05$, Fig. 5D).

373 **Discussion**

374 Using complementary public resources, we characterised immune gene and protein profiles
375 across blood, CSF and peripheral nerve in inflammatory neuropathies. CXCL8 recruitment
376 and complement showed the most consistent effect directions in acute GBS blood, and most
377 selected CSF proteins had positive coefficients in both cohorts. In nerve, immune and repair
378 gene expression was distributed across macrophages, endothelial cells and Schwann cells,
379 with a lower macrophage Fc receptor score in CIDP than in CIAP. Published genetic

380 candidates also showed distinct cellular expression patterns. These findings relate recurring
381 immune themes to the nerve cell populations expressing the relevant genes.

382 Recruitment and complement emerged as recurring expression patterns across different blood
383 preparations in acute GBS. Six modules were significant in the whole blood cohort, while
384 CXCL8 recruitment and complement retained positive effect estimates across all three
385 cohorts and after each cohort was omitted in turn, albeit with wide pooled confidence
386 intervals. This pattern is consistent with experimental evidence implicating chemokines and
387 leukocyte trafficking and with published CSF findings concerning interleukin 8 (IL8)
388 [19,36,45,46,47]. Blood module scores and CSF protein abundance measurements capture
389 different aspects of these processes, making their concordance useful for prioritising
390 pathways to investigate in paired blood and CSF specimens.

391 The CSF findings add protein evidence for inflammation and associated vascular processes.
392 Nine of the ten selected proteins had positive coefficients in both cohorts, and SELE and
393 IL2RA were the two proteins in this subset that also met the source study's adjusted
394 significance threshold in each cohort [36]. Substantial heterogeneity for several proteins
395 indicates that the estimated magnitudes varied considerably between cohorts. Complementary
396 proteomic and peptidomic reports described inflammation, complement and molecular injury
397 [26,33,34,35], broadening the biological context beyond the ten proteins included in the
398 quantitative synthesis.

399 CSF samples the environment surrounding nerve roots and the neuraxis, whereas the BNB
400 regulates exchange across the microvascular endothelium within peripheral nerve. The
401 distribution of endothelial adhesion genes, immune transcripts and Schwann cell repair
402 programmes suggests a plausible connection between immune cell recruitment and local
403 tissue responses. Their temporal relationship remains unresolved because the specimens came
404 from separate cohorts. Paired blood and CSF measurements, spatial nerve profiling and

endothelial permeability assays could help clarify how these processes relate within individual patients.

Within the original analysis of nerve cell groups, the macrophage Fc receptor module showed the strongest statistical evidence of a difference between CIDP and CIAP. The estimated difference remained negative after demographic adjustment and omission analyses, with four receptor or signalling genes contributing to the lower score. Attenuation after composition adjustment indicates that the estimate is sensitive to how macrophage populations are represented in the model, an issue relevant to the diverse macrophage states described in the source atlas [27]. Published findings of complement deposition on endoneurial capillaries and altered peripheral B cell and CD8 effector profiles in CIDP provide broader context for immune involvement at different anatomical sites [29,37]. The lower Fc receptor score therefore characterises this particular comparison with CIAP, with its interpretation depending on macrophage composition and the scope of multiple testing correction.

The nominal positive association between the combined Schwann cell myelin and repair score and disability may reflect changes in the tissue response to injury. Because this module includes both myelin and injury response genes, a higher score does not establish greater repair capacity. The association did not meet the threshold after multiple testing correction ($q = 0.069$), and the small number of donors limits the precision of the estimate. Studies combining gene expression profiling, analysis of tissue organisation and longitudinal clinical assessment could clarify how vascular and Schwann cell responses relate to clinical impairment.

CDH4 expression was concentrated in Schwann cell states and *GNG7* expression in B cells, highlighting populations for further investigation. The reported *CDH4* association was identified in women [32], whereas all CIDP nerve donors were male. Although these expression data inform cellular localisation, potential sex dependent genetic effects remain

430 unresolved. Low expression of *DIRAS1* and *SLC39A3* also limited their precise localisation.
431 Functional studies in the relevant cell populations could help clarify how these genetic
432 candidates relate to peripheral nerve biology.

433 **Limitations**

434 The blood cohorts were small and differed in cell composition and assay platform, resulting
435 in wide confidence intervals for the pooled estimates. The CSF synthesis was restricted to ten
436 proteins selected for reporting in the source publication, while the broader evidence map
437 drew on four selected studies. Both analyses therefore capture only part of the available
438 evidence. Correction across the ten pooled protein tests does not account for selection during
439 the original protein screen. Several CSF estimates were also sensitive to the pooling model.
440 In addition, the pooled CSF tests relied on a normal approximation, whereas the original
441 studies used regression tests in small cohorts.

442 The nerve analysis compared nine donors with CIDP and 11 with axonal neuropathy, limiting
443 interpretation to differences between these disease groups. All CIDP donors were male,
444 recruitment was uneven across centres, and treatment and other clinical differences may have
445 influenced gene expression. The Fc receptor association yielded $q = 0.087$ after correction
446 across all 80 nerve module comparisons, and the effect estimate was attenuated after
447 adjustment for cellular composition. Component and sensitivity analyses were conducted
448 after the initial observation, and subtype estimates were based on few eligible donors. These
449 findings therefore require evaluation in a larger independent nerve cohort.

450 Blood, CSF and nerve data came from separate cohorts, with diagnosis partly linked to
451 specimen type, so disease related differences could not be fully separated from tissue specific
452 variation. The companion spatial panel contained none of the specified Fc receptor module
453 genes, preventing spatial assessment of this association. Genetic estimates were drawn from

454 the published study, whereas our analyses assessed cellular expression. Matched samples and
455 functional experiments are needed to test potential mechanistic links across compartments.

456 **Conclusions**

457 Analysis of public blood, CSF and nerve datasets reveals complementary immune expression
458 patterns in inflammatory neuropathies. CXCL8 related recruitment and complement
459 programmes recur across acute GBS blood cohorts, while CSF protein findings point to
460 inflammatory and vascular processes. The nerve atlas maps immune and repair genes to
461 specific cell populations and identifies a lower macrophage Fc receptor module score in
462 CIDP than in CIAP. The magnitude of this association is sensitive to cellular composition,
463 and its statistical support depends on the scope of multiple testing correction. Together with
464 the cellular localisation of published genetic candidates, these findings identify pathways and
465 nerve cell populations for further investigation in independent cohorts and studies using
466 paired specimens.

467 **List of abbreviations**

468 AIDP, acute inflammatory demyelinating polyradiculoneuropathy. BNB, blood–nerve
469 barrier. CIAP, chronic idiopathic axonal polyneuropathy. CIDP, chronic inflammatory
470 demyelinating polyradiculoneuropathy. CPM, counts per million. CP10k, counts per 10,000.
471 CSF, cerebrospinal fluid. FDR, false discovery rate. GBS, Guillain–Barré syndrome. GWAS,
472 genome-wide association study. INCAT, Inflammatory Neuropathy Cause and Treatment.
473 LIF, leukaemia inhibitory factor. NPX, normalised protein expression. OSM, oncostatin M.
474 PBMC, peripheral blood mononuclear cell.

475 **Declarations**

476 **Ethics approval and consent to participate**

477 This study analysed publicly available deidentified data and published results. The source
478 studies reported their ethics approvals and consent procedures. No new participants were
479 recruited and no new specimens were collected.

480 **Consent for publication**

481 Not applicable. The manuscript contains no identifiable personal information.

482 **Availability of data and materials**

483 Database names, accession identifiers, original filenames and source links are provided in
484 Additional file 1, together with aggregate analysis results. Additional file 2 describes the
485 analysis procedures and the reconstruction of intermediate tables. Analysis code and
486 aggregate results for version 2.3.1 are available in the GitHub repository [48] and are
487 archived on Zenodo [49]. This release includes the CSF coefficient synthesis, macrophage
488 sensitivity analyses, annotation comparisons and source documentation. The donor
489 expression table required by the genetic localisation scripts is supplied in Additional file 4,
490 with instructions for using it with the archived scripts.

491 **Competing interests**

492 The authors declare that they have no competing interests.

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497 **Authors' contributions**

498 YF conceived the study, developed the analytical framework, performed the analyses and
499 drafted the manuscript. FF curated data, validated results and revised the manuscript. SXM
500 contributed to validation, interpretation and manuscript revision. SMF contributed to data
501 curation, literature evaluation and manuscript revision. CJ supervised the methodology and
502 critically revised the manuscript. YH supervised the study, coordinated the project and
503 critically revised the manuscript. All authors read and approved the final manuscript.

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506 **Supplementary information**

507 Additional file 1. Additional file 1 Source Data.xlsx. Aggregate source data and database
508 documentation. This Excel workbook contains 49 worksheets with figure and table source
509 data, sensitivity results and database access records.

510 Additional file 2. Additional file 2 Methods.docx. This Word document describes the
511 analytical procedures, source attribution and reconstruction of intermediate tables.

512 Additional file 3. Additional file 3 Figures.docx. Supplementary Figures 1 to 4. This Word
513 document contains the four supplementary figures and legends on genetic candidate
514 expression, published nerve observations and macrophage sensitivity analyses.

515 Additional file 4. Additional_file_4_Genetic_localisation_input.zip. This archive contains
516 Genetic_donor_celltype.csv, a data dictionary, validation records and instructions for using
517 the donor expression summaries with the version 2.3.1 genetic localisation scripts.

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681 **Tables**

682 **Table 1. Datasets and analytical roles**

Resource	Material	Participants or samples	Analytical role
GSE211225	Whole blood transcriptomics	6 acute GBS, 10 postacute, 6 controls	Primary blood effects for individual participants
GSE31014	Leukocyte microarray	7 GBS, 7 controls	Independent assessment of blood effect direction
PRJNA1293757	PBMC transcriptomics	3 AIDP, 2 controls	Monocyte localisation and assessment of effect direction
PXD002911	CSF proteomics and peptidomics	GBS (AIDP), MS and controls, with pooled discovery material	Published CSF identifications and functional annotations
GSE285983	Sural nerve nuclear transcriptomics	9 CIDP, 11 CIAP in focused comparison	Donor gene module scores, cell fractions and expression localisation
GSE107574	Human endoneurial endothelial transcriptomics	Cultured cells and microvessels	Published BNB expression reference

Resource	Material	Participants or samples	Analytical role
Kmezic et al. 2023	CSF Olink proteomics	Discovery cohort with 6 GBS and 20 controls. Replication cohort with 6 GBS and 6 controls.	Synthesis of ten paired CSF protein coefficients from source Table 4B
PXD056286	Sural nerve histology and proteomics	55 CIDP biopsies for histology. Proteomics in 9 CIDP cases and 2 controls.	Published tissue observations and calculated Wilson interval

683 GBS, Guillain–Barré syndrome. AIDP, acute inflammatory demyelinating polyradiculoneuropathy.

684 CIDP, chronic inflammatory demyelinating polyradiculoneuropathy. CIAP, chronic idiopathic axonal

685 polyneuropathy. CSF, cerebrospinal fluid. MS, multiple sclerosis. PBMC, peripheral blood

686 mononuclear cell. BNB, blood–nerve barrier. Eligible nerve donor numbers vary by cell group.

687 **Table 2. Pooled effects of acute GBS blood modules**

Gene module	Hedges g	95% CI	P value	I ² (%)	Cohorts
CXCL8 signalling via CXCR1 and CXCR2	1.075	−0.634 to 2.785	0.114	0.0	3
LIF and OSM signalling via gp130	0.271	−3.978 to 4.519	0.810	81.2	3
Complement	1.111	−1.728 to 3.950	0.234	57.2	3
Fc receptor	0.288	−5.510 to 6.086	0.851	72.7	3
Transendothelial migration	0.694	−0.958 to 2.347	0.212	1.1	3
Interferon signalling via JAK and STAT	0.677	−0.943 to 2.297	0.214	0.0	3
Inflammatory monocyte	1.026	−1.702 to 3.754	0.247	49.5	3

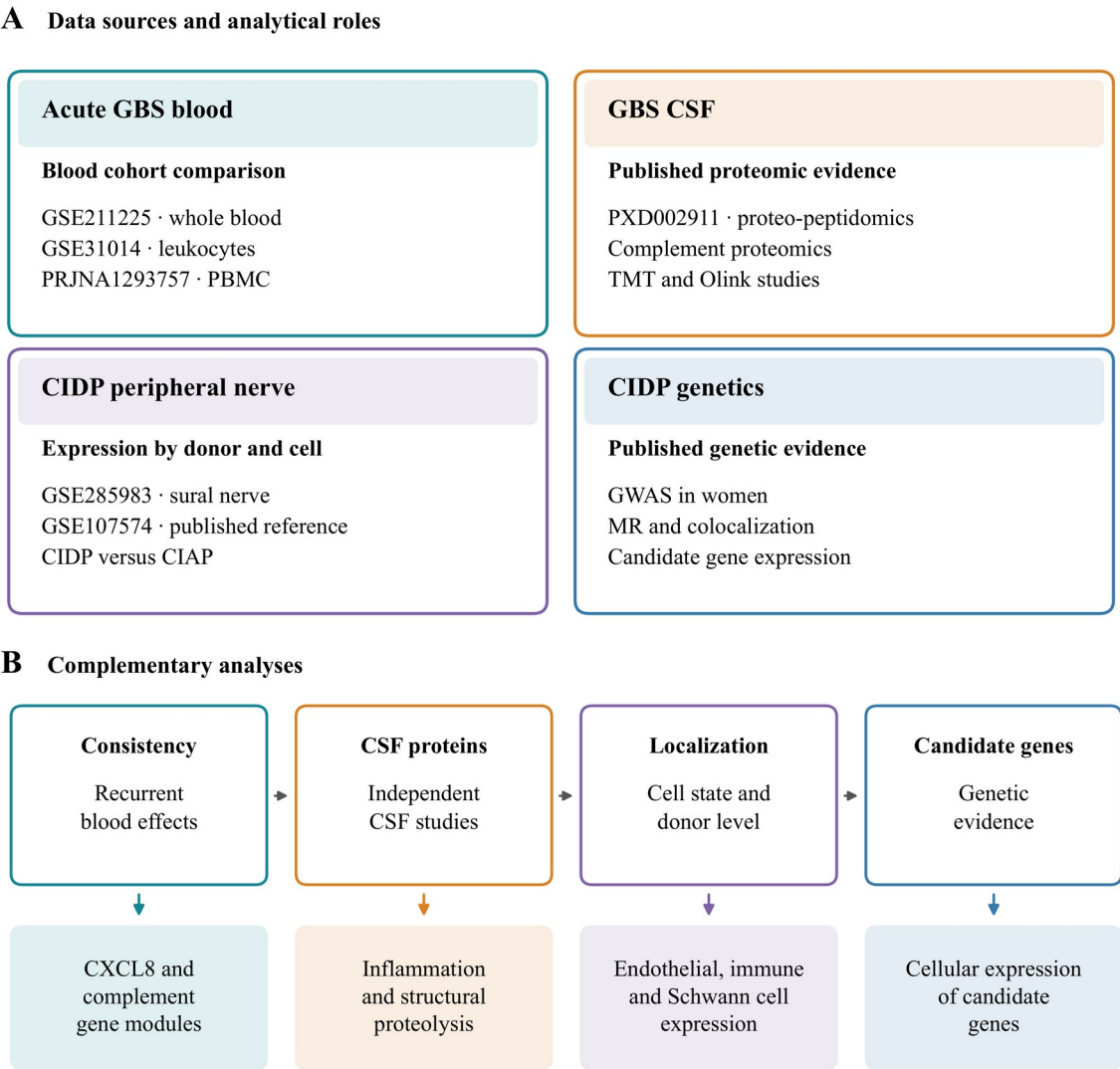
688 Positive Hedges g indicates a higher score in acute GBS. Variance between cohorts was estimated by
689 the DerSimonian and Laird method. Confidence intervals and P values use the modified Hartung and
690 Knapp procedure, with residual scale bounded below by one. Each estimate combines three cohorts.
691 CI, confidence interval.

692 **Table 3. Evidence across blood, CSF and peripheral nerve**

Programme	Blood evidence	CSF evidence	Peripheral nerve evidence	Interpretation
CXCL8 recruitment	Positive in 3 cohorts. Pooled P = 0.114. Whole blood q < 0.05	Published innate immune and inflammatory protein findings	Myeloid localisation and endothelial trafficking context	Recurring recruitment expression
Complement	Positive in 3 cohorts. Pooled P = 0.234. Whole blood q < 0.01	Published complement protein findings	Macrophage localisation	Complement gene and protein findings
Fc receptor	Whole blood q < 0.05, with mixed directions across cohorts	Not directly evaluated	Macrophage q = 0.0087 within eight modules and q = 0.087 across 80 nerve comparisons	Lower macrophage score, attenuated after composition adjustment
Interferon response	Whole blood q < 0.05, with mixed directions across cohorts	Published evidence of antiviral defence	Published <i>OAS1</i> enrichment in macrophages and <i>MX1</i> enrichment in arterial endothelial cells. Published lower <i>MX1</i> in nonmyelinating Schwann cells in PNP versus controls [27].	Published gene level expression context
Endothelial adhesion	Migration module q = 0.126 in whole blood. Positive in whole blood and leukocytes, negative in monocytes	Not directly evaluated	<i>ICAM1</i> and <i>VCAM1</i> localisation to BNB endothelial cells	Endothelial expression of adhesion genes
Schwann repair	Not evaluated	Published evidence of axonal domain peptide degradation	Repair and stress programmes localised to Schwann cells	Structural injury and Schwann cell repair context

693 Blood comparisons concern acute GBS and healthy controls. Nerve comparisons in this study concern
694 CIDP and CIAP. Blood q values refer to seven module tests within the whole blood cohort, and nerve
695 q values refer to the stated testing families. The nerve interferon entry uses published marker results
696 from the full source atlas and a published PNP versus control comparison [27]. A dedicated nerve
697 interferon module was not tested in the CIDP versus CIAP analysis. The evidence types do not form a
698 common scale across compartments. PNP, polyneuropathy.

700 **Figure 1. Study design and analytical roles**



701 Public blood, CSF, peripheral nerve and genetic resources contributed complementary measurements.

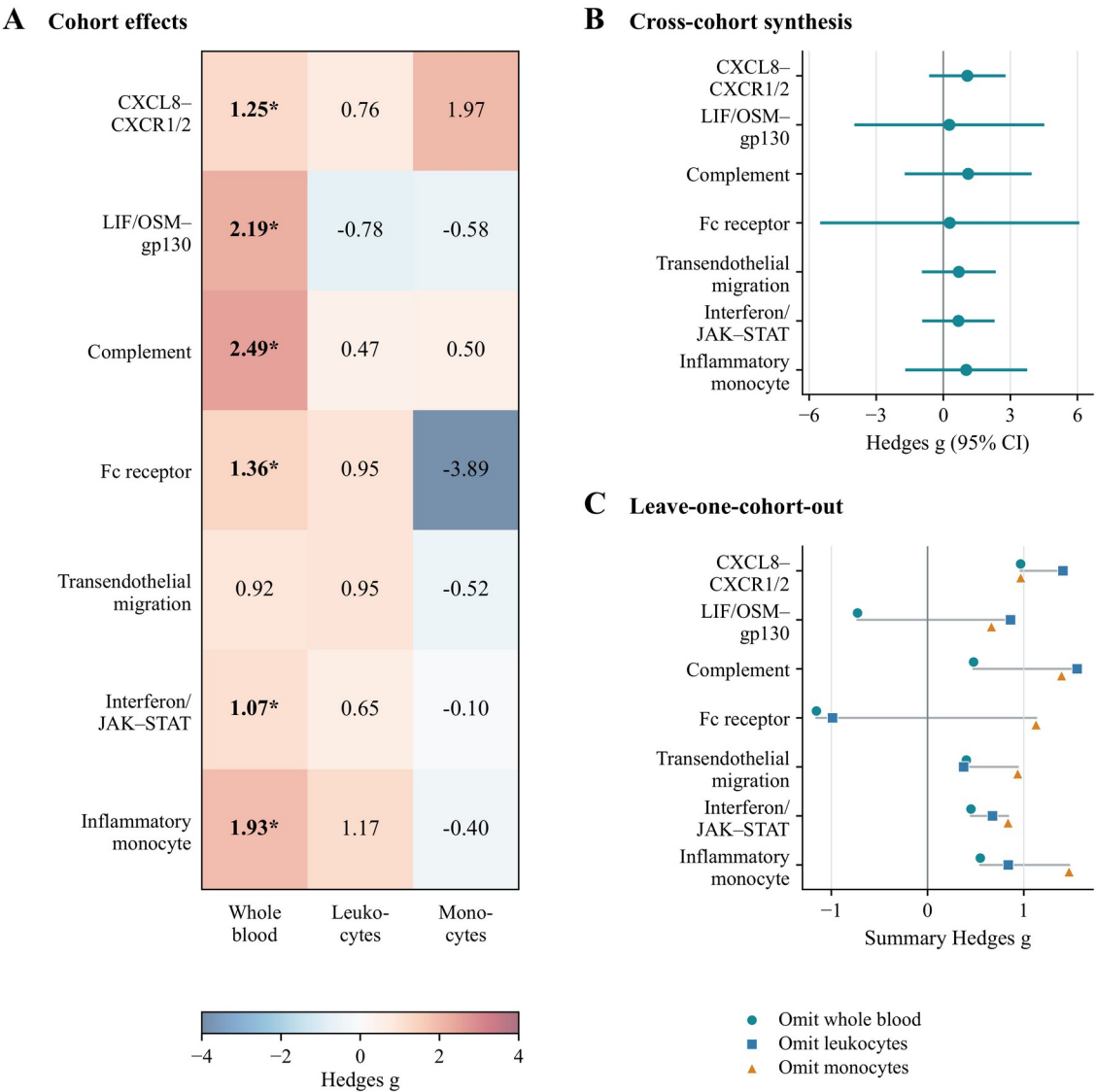
702 Blood cohorts provided gene module effects, CSF reports provided protein and peptide findings, and

703 the nerve atlas provided expression by donor and cell group. Published genetic associations guided

704 candidate selection for nerve expression analysis. Arrows summarise analytical connections and

705 biological questions.

706 **Figure 2. Acute GBS blood modules across cohorts**



707 (A) Hedges g for seven gene modules in whole blood, leukocytes and monocyte pseudobulk.

708 Asterisks indicate $q < 0.05$ after correction within each cohort. (B) Pooled estimates from models with

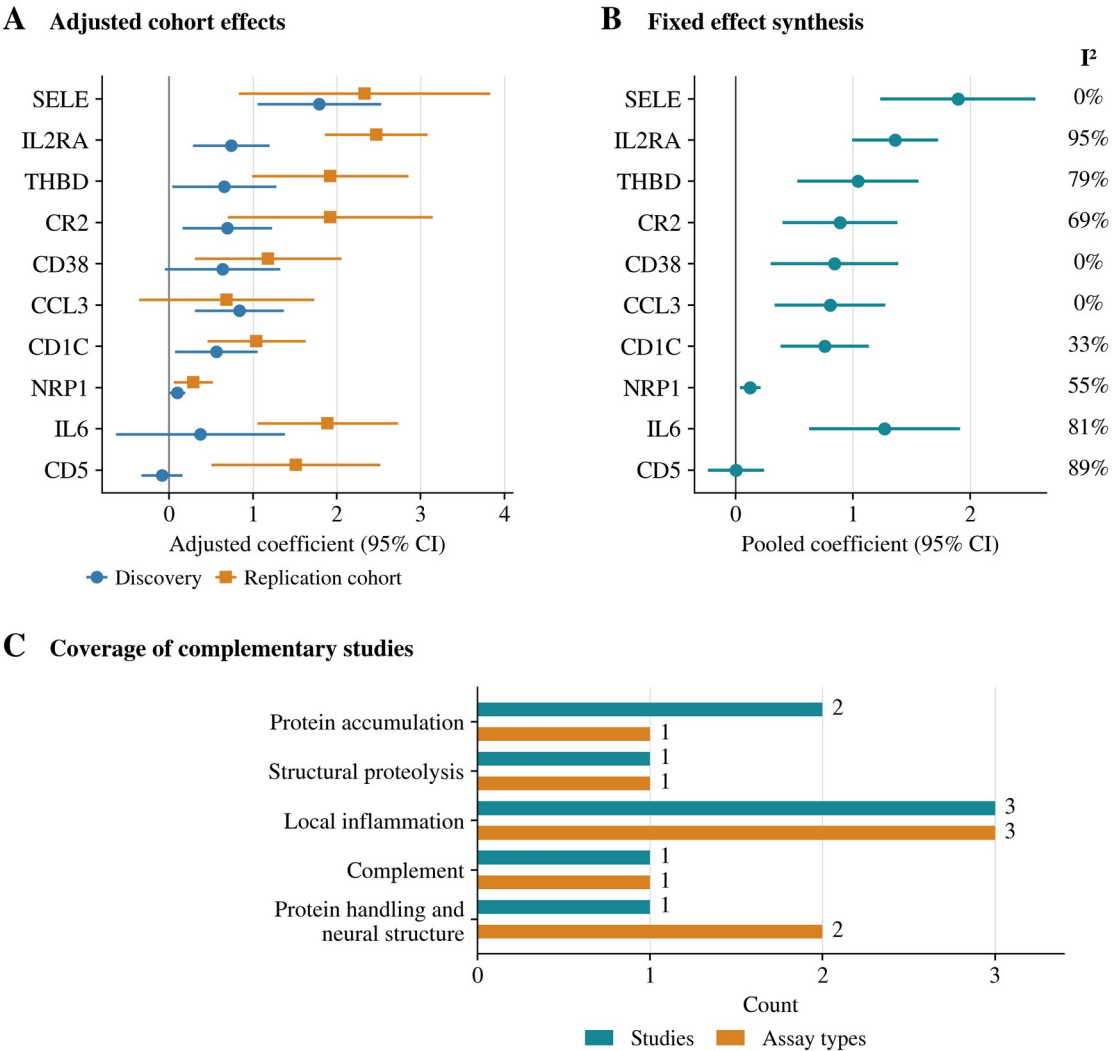
709 random effects and 95% confidence intervals calculated using the modified Hartung and Knapp

710 method. (C) Pooled estimates after each cohort was omitted in turn. CXCL8 recruitment and

711 complement retained positive estimates in all subsets, and both effects in each retained pair were

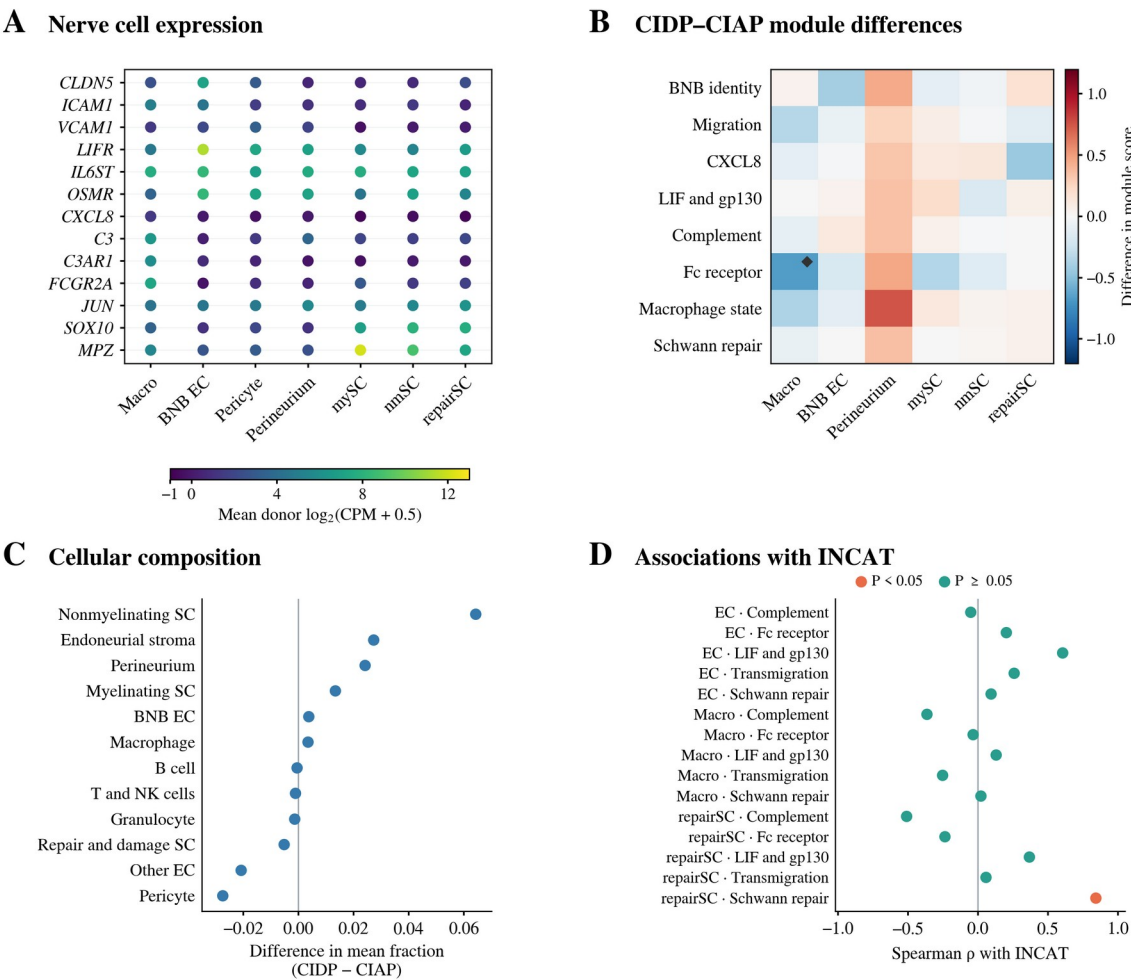
712 positive. All pooled intervals in B include zero.

713 **Figure 3. Synthesis of published CSF protein coefficients**



714 (A) Coefficients for GBS versus healthy controls from the discovery and replication cohorts of
715 Kmezcic et al., expressed on the log2 NPX scale and adjusted for age, sex and the CSF sample
716 handling marker CCL19 [36]. Error bars show 95% confidence intervals calculated from the reported
717 standard errors using a normal approximation. (B) Estimates pooled with inverse variance weights in
718 a model with a fixed effect. The analysis includes the ten proteins with paired coefficients in source
719 Table 4B, with IL8 considered separately. Nine pooled tests have $q < 0.05$, while CD5 has $q = 0.984$.
720 I^2 describes heterogeneity. The source study found adjusted significance in both cohorts for SELE and
721 IL2RA. (C) Coverage by study and assay type across four descriptive reports [26,33,34,35].
722 Concentration measurements are grouped as one broad assay type.

723 **Figure 4. Immune and repair modules in CIDP nerve**



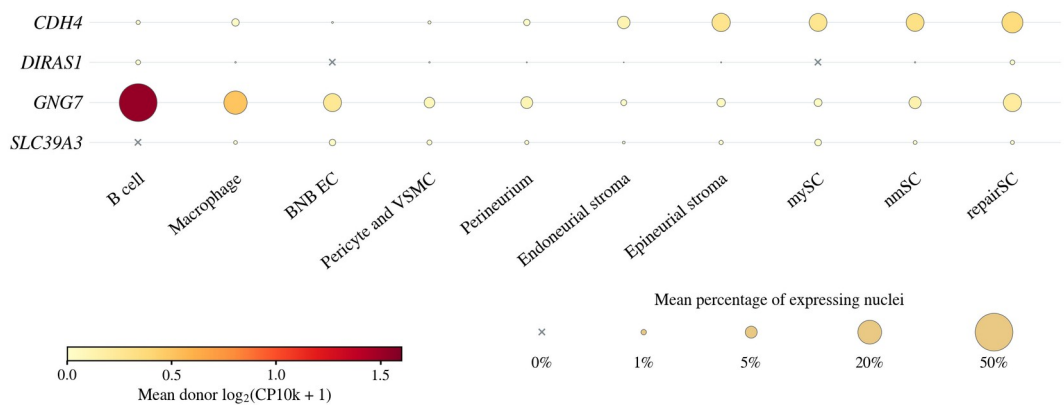
724 (A) Mean donor gene expression across seven CIDP nerve cell groups. Colour shows the arithmetic
725 mean of donor $\log_2(\text{CPM} + 0.5)$ values on a common scale across genes and cell groups. All dots
726 have the same area. Each cell group includes nine donors. Granulocytes are omitted because matching
727 expression summaries were unavailable. (B) Differences in donor module scores between CIDP and
728 CIAP for six selected cell groups. Diamonds indicate $q < 0.05$ after correction within a cell group.
729 The macrophage Fc receptor result has $q = 0.087$ after correction across all 80 eligible comparisons.
730 (C) Differences in mean cell fractions, calculated as CIDP minus CIAP. The comparisons in B and C
731 use the Mann and Whitney U test with donors as independent samples. (D) Spearman correlations
732 between donor module scores and INCAT disability. Orange indicates $P < 0.05$ and teal indicates $P \geq$
733 0.05 . All disability correlations have $q > 0.05$ after correction. BNB EC denotes endoneurial
734 microvascular endothelial cells. Supplementary Figures 3 and 4 present the macrophage sensitivity
735 analyses. Macro denotes macrophages. mySC denotes myelinating Schwann cells, nmSC denotes
736 nonmyelinating Schwann cells, and repairSC denotes Schwann cells annotated as repair and damage
737 states.

738 **Figure 5. Evidence across compartments and expression of genetic candidates**

A Evidence types across compartments

Programme	Blood	CSF	Peripheral nerve
CXCL8 recruitment	Whole blood test q = 0.045 Pooled P = 0.114	Published findings Innate immune and inflammatory proteins	Expression context Myeloid genes and endothelial trafficking
Complement	Whole blood test q = 0.0076 Pooled P = 0.234	Published findings Complement proteins	Expression localisation Macrophage complement genes
Fc receptor	Whole blood test q = 0.038 Mixed cohort directions	Not assessed	Macrophage test q = 0.0087 within 8 modules q = 0.087 across 80 tests
Interferon response	Whole blood test q = 0.045 Mixed cohort directions	Published annotation Antiviral defence	Published atlas markers [27] OAS1 and MX1 localisation Published PNP versus control [27] Lower MX1 in nmSC
Endothelial adhesion	Migration module test Whole blood q = 0.126 Mixed cohort directions	Not assessed	Expression localisation ICAM1 and VCAM1 in BNB EC
Schwann repair	Not assessed	Published findings Axonal domain peptide degradation	Expression localisation Schwann repair and stress genes

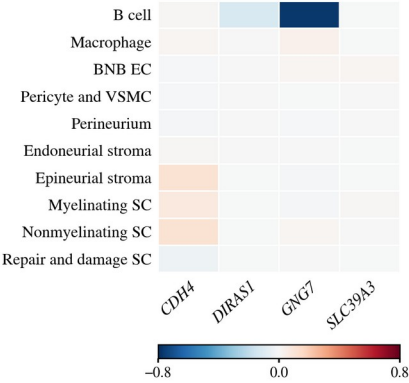
B CIDP nerve expression of genetic candidates



C Published genetic evidence

<i>CDH4</i>	GWAS in women OR 4.37
<i>DIRAS1</i>	MR and colocalisation β -0.45 PP.H4 0.94
<i>GNG7</i>	MR and colocalisation β -2.11 PP.H4 0.93
<i>SLC39A3</i>	MR and colocalisation β -1.42 PP.H4 0.87

D CIDP-CIAP expression



739 (A) Descriptive evidence map with the evidence type stated in each cell. The categories are not
 740 ranked. Blood q values refer to seven module tests within the whole blood cohort. The nerve
 741 macrophage Fc receptor q values refer separately to eight modules within macrophages and all 80
 742 eligible nerve module comparisons. All pooled blood module effects have $P > 0.05$. Published nerve
 743 interferon evidence comes from Supplementary Data 3 and 9 of Heming et al. [27]. The marker
 744 results identify *OAS1* enrichment in macrophages and *MX1* enrichment in arterial endothelial cells
 745 relative to other atlas nuclei. The published disease comparison shows lower *MX1* in nonmyelinating
 746 Schwann cells in polyneuropathy than in source controls. (B) Expression of *CDH4*, *DIRAS1*, *GNG7*
 747 and *SLC39A3* across CIDP nerve cell groups. Dot area is proportional to the arithmetic mean of donor
 748 percentages of expressing nuclei. Colour shows the arithmetic mean of donor mean $\log_2(\text{CP10k} + 1)$
 749 values. Both summaries give equal weight to each of the nine CIDP donors. Crosses indicate zero
 750 detected expression. (C) Published genetic estimates from Du et al. that guided candidate selection
 751 [32]. The *CDH4* estimate is for rs80339443 in women. MR, Mendelian randomisation. PP.H4,
 752 posterior probability of a shared causal variant. (D) Differences in donor mean expression between
 753 CIDP and CIAP. Red indicates higher expression in CIDP and blue indicates lower expression. All
 754 displayed comparisons have $q > 0.05$. PNP denotes polyneuropathy. BNB EC denotes endoneurial
 755 microvascular endothelial cells. VSMC denotes vascular smooth muscle cells. mySC denotes
 756 myelinating Schwann cells, nmSC denotes nonmyelinating Schwann cells, and repairSC denotes
 757 Schwann cells annotated as repair and damage states.